BCA-301: Mathematics-III (Important Q&A)

Weightage: 70 (Block I: 30, Block II: 40) Marks

Q1. Explain the Limit and Continuity of a Complex Function.

A complex function f(z) has a limit L as z approaches z0 if the value of f(z) can be made arbitrarily close to L. It is continuous at z0 if the limit exists and equals f(z0). Both real and imaginary parts must be continuous.

Mathematical/Theoretical Expression: $\lim(z \to z_0)\ f(z) = f(z_0)$

Q2. Define Differentiability and Analyticity of a complex function.

A function is differentiable at a point if its derivative exists there. It is Analytic (or holomorphic) at a point if it is differentiable at that point AND in some neighborhood around it.

Mathematical/Theoretical Expression: $f'(z) = \lim(\Delta z \to 0)\ [f(z+\Delta z) - f(z)] / \Delta z$

Q3. State the Cauchy-Riemann (C-R) Equations.

The C-R equations are a set of two partial differential equations which are necessary conditions for a complex function f(z) = u(x,y) + iv(x,y) to be analytic.

Mathematical/Theoretical Expression: $\partial u/\partial x = \partial v/\partial y$ AND $\partial u/\partial y = -\partial v/\partial x$

Q4. What are Laplace Equations in complex variables?

If a complex function is analytic, its real (u) and imaginary (v) parts satisfy Laplace's equation. Such functions are called harmonic functions.

Mathematical/Theoretical Expression: $\nabla^2 u = \partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = 0$

Q5. Explain Cauchy's Integral Formula.

This fundamental theorem states that if a function f(z) is analytic inside and on a simple closed contour C, the value of f(z) at any point z0 inside C is determined by its values on the boundary.

Mathematical/Theoretical Expression: $f(z_0) = (1 / 2\pi i) \oint [f(z) / (z - z_0)] dz$

Q6. Define Infinite Sequences and Series of complex numbers.

A sequence is an ordered list of complex numbers. A series is the sum of the elements of a sequence. A series converges if the sequence of its partial sums approaches a finite limit.

Mathematical/Theoretical Expression: $S_n = \Sigma(k=1 \text{ to } n) z_k$

Q7. What is an Improper Integral?

An improper integral is a definite integral where either one or both of the limits are infinite, or the integrand approaches infinity at one or more points in the range of integration.

Mathematical/Theoretical Expression: $\int[a \text{ to } \infty] f(x) dx = \lim(b\to\infty) \int[a \text{ to } b] f(x) dx$

Q8. Explain Power Series and Radius of Convergence.

A power series is an infinite series of the form $\Sigma a_n(z - z_0)^n$. The radius of convergence (R) is the distance within which the power series converges absolutely.

Mathematical/Theoretical Expression: $R = \lim(n\to\infty) |a_n / a_{n+1}|$

Q9. How are Power Series used to solve Ordinary Differential Equations?

The power series method involves assuming a solution of the form $y = \Sigma a_n x^n$, substituting it into the differential equation, and finding a recurrence relation for the coefficients a_n.

Mathematical/Theoretical Expression: $y(x) = a_0 + a_1 x + a_2 x^2 + \ldots$

Q10. Define Legendre's Equation and Polynomials.

Legendre's differential equation frequently arises in physics (like solving Laplace's equation in spherical coordinates). The solutions are called Legendre Polynomials, denoted as $P_n(x)$.

Mathematical/Theoretical Expression: $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$

Q11. What are Bessel Equations and Bessel Functions?

Bessel's equation arises when finding separable solutions to Laplace's equation in cylindrical coordinates. Solutions are Bessel functions of the first kind (J_n) and second kind (Y_n).

Mathematical/Theoretical Expression: $x^2y'' + xy' + (x^2 - n^2)y = 0$

Q12. Explain the concept of Laplace Transforms.

The Laplace Transform is an integral transform that converts a function of a real variable t (often time) to a function of a complex variable s (frequency). It transforms differential equations into algebraic equations.

Mathematical/Theoretical Expression: L{f(t)} = F(s) = ∫[0 to ∞] e^(-st) f(t) dt

Q13. What is the Inverse Laplace Transform?

The Inverse Laplace Transform converts a function from the complex s-domain back into the time domain (t). It is used to find the solution of the original differential equation.

Mathematical/Theoretical Expression: $L^{-1}\{F(s)\} = f(t)$

Q14. State the First Shifting Theorem (s-axis shift).

If the Laplace transform of f(t) is F(s), then multiplying f(t) by an exponential e^(at) shifts the transform on the s-axis by 'a'.

Mathematical/Theoretical Expression: L{e^(at) * f(t)} = F(s - a)

Q15. State the Second Shifting Theorem (t-axis shift).

If a function is shifted in time by 'a' and multiplied by a unit step function u(t-a), its Laplace transform is multiplied by an exponential factor e^(-as).

Mathematical/Theoretical Expression: L{f(t-a) * u(t-a)} = e^(-as) * F(s)

Q16. How do you find the Laplace Transform of a Derivative?

The Laplace transform of a derivative relies on the transform of the function itself and its initial conditions, which makes it perfect for solving Initial Value Problems.

Mathematical/Theoretical Expression: L{f'(t)} = sF(s) - f(0)

Q17. What is the Cauchy-Riemann equation in polar coordinates?

For a function represented in polar form f(r, θ) = u(r, θ) + iv(r, θ), the C-R equations adapt to polar variables, essential for circular domain problems.

Mathematical/Theoretical Expression: $\partial u/\partial r = (1/r)\partial v/\partial \theta$ and $\partial v/\partial r = -(1/r)\partial u/\partial \theta$

Q18. What is an Essential Singularity in Complex Variables?

A singularity z0 is essential if the principal part of the Laurent series of f(z) has an infinite number of non-zero terms. The function's behavior near this point is highly erratic.

Mathematical/Theoretical Expression: e.g., $f(z) = e^{(1/z)}$ at $z = 0$

Q19. Explain Absolute vs. Conditional Convergence of Series.

A series converges absolutely if the sum of the absolute values of its terms converges. It converges conditionally if the series converges but the sum of absolute values diverges.

Mathematical/Theoretical Expression: $\Sigma|z_n|$ converges $\Rightarrow \Sigma z_n$ converges absolutely

Q20. What is the Convolution Theorem for Laplace Transforms?

The convolution theorem states that the Laplace transform of the convolution of two functions is the product of their individual Laplace transforms.

Mathematical/Theoretical Expression: $L\{f(t) * g(t)\} = F(s) * G(s)$